MIL-S-11183F
31 December 1985
SUPERSEDING
MIL-S-11183E
2 June 1977

MILITARY SPECIFICATION

SCALES, DIAL AND BEAM INDICATING, BAKER'S, DOUGH PIECE CHECKING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers baker's weighing scales for checking the weight of a dough piece.

1.2 Classification. The scales covered by this document shall be of the following types and classes, as specified (see 6.2):

Type I - Over and under scale (see figure 1)

Class 1 - Avoirdupois, 3-pound capacity

Class 2 - Metric, 1500-gram capacity

Type II - Even-balance scale, 5-pound capacity (see figure 2)

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6670

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SPECIFICATION

FEDERAL

PPP-B-1122 - Balances, Scales, and Accessories, Packaging and Packing Of

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

BAKING INDUSTRY SANITATION STANDARDS COMMITTEE (BISSC)

Sanitation Standard No. 23 for Floor Scales and Ingredient Scales

(Application for copies of BISSC publications should be addressed to Baking Industry Sanitation Standards Committee, 521 Fifth Ave., New York, NY 10017.)

NATIONAL BUREAU OF STANDARDS (NBS)

Handbook 44 - Specification, Tolerances, and Other Technical Requirements For Commercial Weighing and Measuring Devices

(Application for copies of Handbook 44 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Standard product. The scales delivered under this document shall be the manufacturer's commercial product, except for any changes necessary to comply with document requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.2 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial scales, provided the completed item complies with all provisions of this document. Recycled material may be used (see 6.3).

3.3 Standards. In addition to the requirements specified herein, the scales and weights shall meet the applicable requirements as specified in NBS Handbook 44 and the applicable requirements of the BISSC Standard No. 23.

3.3.1 Compliance. Prior to commencing delivery, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the scale he proposes to furnish under this document meets the requirements of NBS Handbook 44 and BISSC Standard No. 23 (see 4.2.4).

3.4 Design and construction.

3.4.1 Type I. The type I scale shall be a bench scale with a dial indicator and a tare beam, similar to figure 1, and shall conform to the dimensions specified in table I. The scale shall be designed to show an over or under weight indication on the dial face by a balance indicator when objects are placed on the load receiver. The scale shall have a platform load receiver and a weight platter for removable weights designed to counterbalance the weight of objects being weighed on the platform load receiver.

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TABLE I. Dimensions, type I scales

Dimension	Minimum (inches)	Maximum (inches)
Overall		
Length	15 1/2	19 1/2
Width	5	7 1/2
Height	15	17 1/2
Load receiver		
Width	6	8 1/2
Length	7	9

3.4.1.1 Class 1. The class 1 scale shall be designed to weigh in the avoirdupois weight system and shall have a minimum capacity of 3 pounds. The tare beam for the scale shall have a capacity of 1 pound with maximum graduations of 1/4 ounce. The dial indicator chart shall have an over and under indication of 1 ounce with 1/4-ounce graduations.

3.4.1.2 Class 2. The class 2 scale shall be designed to weigh in the metric system and shall have a minimum capacity of 1500 grams. The tare beam shall have a capacity of 500 grams with maximum graduations of 5 grams. The dial indicator chart shall have an over and under indication of 25 or 30 grams with 5-gram graduations.

3.4.1.3 Dial indicator. The dial indicator shall be readable from both ends of the scale and shall be enclosed in a metal housing. There shall be a clear enclosure over the chart. The dial indicator chart shall be marked to indicate the zero position and the over and under indication.

3.4.1.4 Tare beam. The tare beam shall be notched and shall have a poise. The poise shall not be detachable from the tare beam without the use of tools.

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3.4.1.5 Damping device. The scale shall have a hydraulic damping device, and the device shall be accessible for adjusting for temperature changes or for adjusting damping oscillation.

3.4.1.6 Load receiver and weight receiver. The load receiver and weight receiver shall be constructed of corrosion-resisting material. There shall be straight or flanged sides on the load receiver and weight receiver in accordance with the manufacturer's current commercial practice.

3.4.2 Type II scale. The type II scale shall be an even-balance bench scale with two 9-inch flat corrosion-resisting steel plates, a tare beam, and balance weights. The scale shall be similar to figure 2. One plate shall be used to hold the balance weights and the other plate to hold the item to be weighed. The scale shall be designed to weigh in the avoirdupois system and shall have a minimum capacity of 5 pounds. The tare beam shall have a capacity of 1 pound with maximum graduations of 1/4 ounce. The beam shall be notched and shall contain a nondetachable poise. The scale shall be constructed with pivots and bearings in accordance with the manufacturer's current commercial practice. The overall nominal dimensions shall be 20 inches long, 8 inches high, and 10 inches deep.

* 3.5 Weights and weight box. Weights shall be furnished with the scale in accordance with table II. Weights, if not constructed of corrosion-resisting material, shall be coated to resist corrosion. All weights shall be marked with the weight value. There shall be a wooden or durable plastic box to contain the weights.

TABLE II. Weights for scales

Scale	Quantity	Weights
Type I		
Class 1		
3 pounds	2	1 pound
Class 2		
1500 grams	1	500 grams
	2	200 grams
	1	100 grams
Type II		
5 pounds	1	2 pounds
	2	1 pound

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3.6 Performance.

3.6.1 Overload. With type I or type II scale supporting a load equal to 125 percent of its rated capacity, the scale shall evidence no permanent deformation, bends, breakage, or separation of parts, and the pointer shall return to the initial zero position after the load is removed, when tested as specified in 4.3.1.

3.6.2 Tolerances. The increasing load and shift tolerance for the scales shall be as specified in NBS Handbook 44 for the acceptance tolerance for small capacity scales (see 4.3.2.1 and 4.3.2.2).

3.7 Finish. The scale shall be finished in accordance with the manufacturer's current commercial practice.

3.8 Markings for identification. The scales shall be marked with the manufacturer's name or readily identifiable trademark and model number.

3.9 Repair and maintenance. Provisions shall be made to allow for weight adjustment, service, and replacement of components without disturbing or disassembly of any major components. In addition, there shall be free access to inspect, service, and adjust balance of tare beam and dial indicator, when applicable, without requiring disassembly of any component on the unit.

3.10 Workmanship. The finished scale shall have no burrs, slivers, splinters, or broken, loose, or missing fasteners.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item inspection. A lot shall consist of all of the scales of one type and class offered for inspection at one time. The sample unit shall be one completely fabricated scale.

* 4.2.2.1 Visual examination. The end item shall be examined for the defects in table III. The inspection level shall be II, and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE III. Classification of defects

Examine	Defect	Classification	
		Major	Minor
Finish	Not manufacturer's current commercial finish		201
Construction and workmanship (applicable to all components)	Any component not accessible for servicing without disassembly of other components	101	
	Burr, sliver, or splinter	102	
	Broken, loose, or missing fasteners		202
	Platform load receiver or weight receiver missing		203
Capacity	Not capacity specified	103	
Tare bar	Not capacity and maximum graduations specified where applicable	104	
	Not notched	105	
	Poise detachable		204

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TABLE III. Classification of defects - Continued

Examine	Defect	Classification	
		Major	Minor
Dial indicator	Not readable from both sides of the scale	106	
	Not marked as specified	107	
Damping device (type I)	Missing	108	
	Not adjustable or accessible	109	
Weights	Missing or not consistent with capacity	110	
Marking (identification)	Missing, incomplete, or not legible		205

4.2.2.2 Dimensional examination. Inspection shall be made of the end item for compliance with specified dimensions and as specified in NBS Handbook 44. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2, and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.2.2.3 End item testing. Sample scales shall be tested as specified in 4.3.1 through 4.3.2.2. Failure of any test shall be recorded as a defect. The inspection level shall be S-2, and the AQL shall be 2.5, expressed as defects per hundred units.

4.2.3 Packaging inspection. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with PPP-B-1122.

4.2.4 Standard compliance. Acceptable evidence of meeting the requirements of BISSC Standard No. 23 shall be a copy of the currently valid, serially numbered Certificate of Authorization issued by the Baking Industry Sanitation Standards Committee. Proof of compliance with the requirements of 3.3.1 shall be available to the contracting officer.

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4.3 Tests.

4.3.1 Overload test. The type I or type II scale shall have the poise on the tare beam set for zero and a load equal to 125 percent of the capacity of the scale applied to the load receiver. Equal loads shall be applied to the weight receiver of platform as applicable. The scales shall be examined before and after removing the load to determine compliance with the requirements specified in 3.6.1.

4.3.2 Tolerance tests.

4.3.2.1 Increasing loads test. The tolerance of the scale at increasing loads shall be determined at 1/8, 1/4, 1/2, and full capacity of the scale. Place a weight equal to each of these loads on the center of the platform and observe the scale to determine compliance with the tolerance requirements specified in 3.6.2.

4.3.2.2 Shift test. The shift tolerance shall be determined by placing a load equal to 1/4 capacity of the scale successively at four points on the load receiver equidistant from the center of the platform and the front, rear, left, and right side of the platform. The scale shall be observed to determine compliance with the requirements specified in 3.6.2.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking for level A, B, or C shall be in accordance with the applicable provisions of PPP-B-1122 (see 6.2).

6. NOTES

6.1 Intended use. The scales are intended primarily to be used by bakers to check the weight of dough pieces leaving the divider and before entering the proofer. The scale can also be used for weighing of dough on a bench. The type II scale is to be used primarily with the portable bakery covered by MIL-B-43322.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type and class of scale required (see 1.2).
c. Selection of the applicable levels of preservation, packing, and marking (see 5.1).

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* 6.3 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.2).

6.4 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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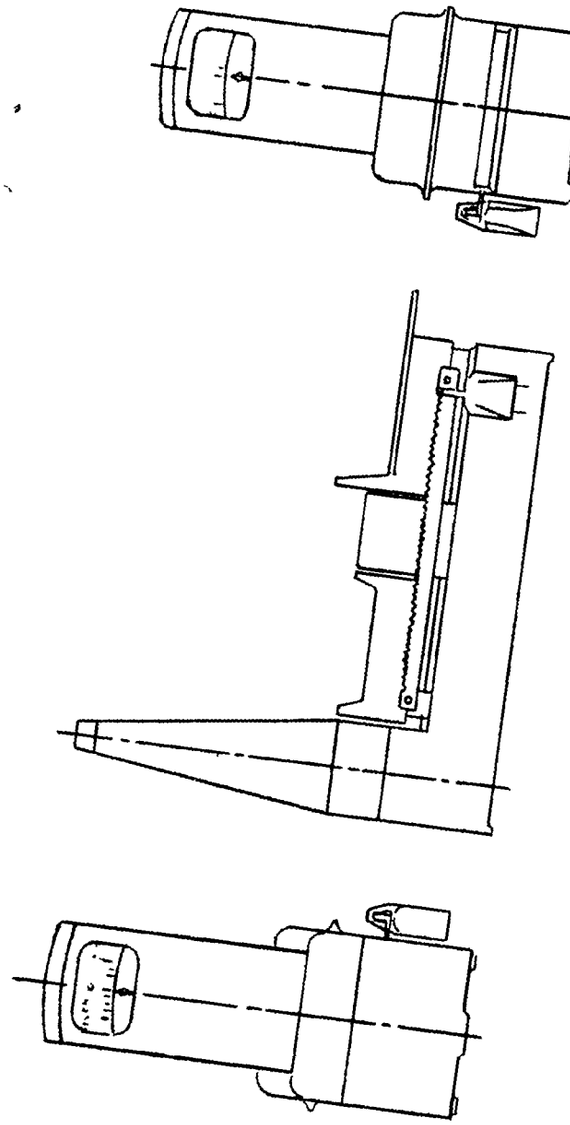


Figure 1. Type I, Over and Under Scale

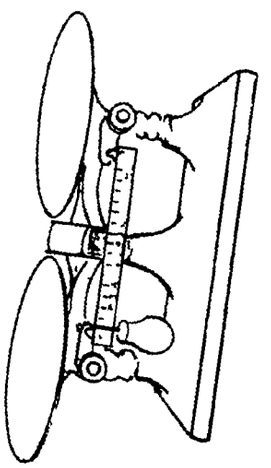


Figure 2. Type II, Even-Balance Scale

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-11183F	SCALES, DIAL AND BEAM INDICATING, BAKER'S DOUGH PIECE CHECKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)* _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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